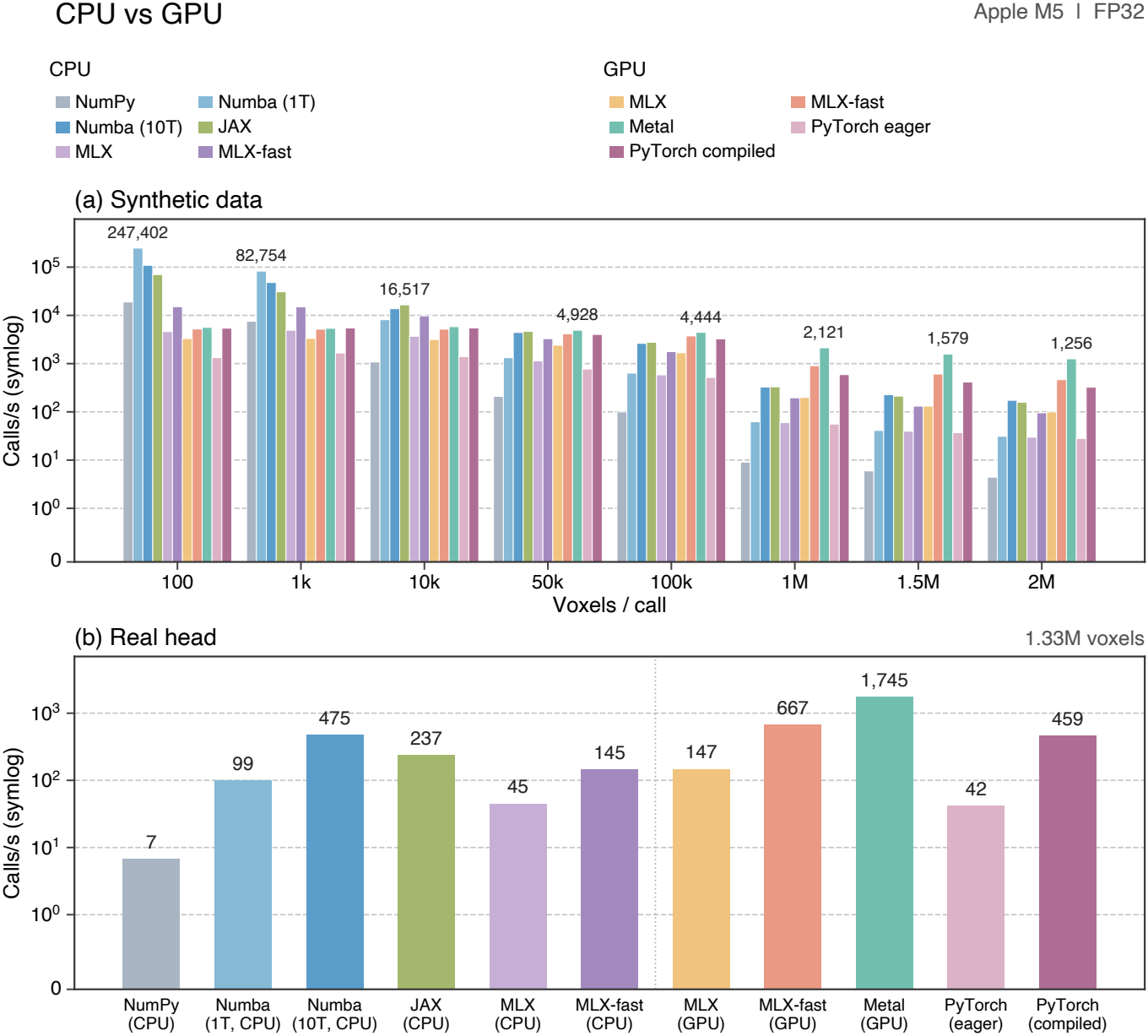
CPU vs GPU
Apple M5 | FP32
CPU
NumPy
Numba (1T)
Numba (10T)
JAX
MLX
MLX-fast
GPU
MLX
MLX-fast
Metal
PyTorch eager
PyTorch compiled
(a) Synthetic data
Calls/s (symlog)
247,402
82,754
16,517
4,928
4,444
2,121
1,579
1,256
100
1k
10k
50k
100k
1M
1.5M
2M
Voxels / call
(b) Real head
1.33M voxels
Calls/s (symlog)
7
99
475
237
45
145
147
667
1,745
42
459
NumPy (CPU)
Numba (1T, CPU)
Numba (10T, CPU)
JAX (CPU)
MLX (CPU)
MLX-fast (CPU)
MLX (GPU)
MLX-fast (GPU)
Metal (GPU)
PyTorch (eager)
PyTorch (compiled)